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NMR studies of block copolymer micelles

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ABSTRACT

Block copolymers dissolved in selective solvent often self-assemble. We review the use of NMR to study this process and to characterize the aggregates formed. We stress the need to consider the polydispersity of the polymers and the fact that the increase in NMR relaxation rates observed upon aggregation can be assigned to contributions from the block copolymer micelle tumbling.

1. Introduction

Block copolymers are made from two (or more) different monomeric species forming distinct sections in the molecular architecture. When they are dissolved in a selective solvent, which is a good solvent for one of the blocks but poor for the other, they may self-assemble into aggregates. This illustrates the general principle that insolubility of one part of a molecule carrying distinct parts in a solvent leads to aggregation; this is most clearly seen in amphiphilic low-molecular weight surfactants. Hence the aggregates formed are often referred to as block copolymer micelles. Different block copolymers show solvent selectivity towards a number of different solvents, from water to non-polar solvents. In addition, by varying the polymer architecture (monomer type, block size, ratio of block sizes and molecular weight) as well as by adding additional components, and by varying concentrations, temperature etc, it is possible to tune the properties of the block copolymer micelles in great detail, which make them amenable to a number of different applications. These include detergency, cosmetics and pharmaceutical applications.

For a long time NMR has been used to study surfactant aggregation, where it has contributed significantly to our present understanding. In the case of block copolymer micelles, various NMR approaches are increasingly used and it is safe to assume that the use and importance of NMR will continue to increase.

This review deals with NMR studies of block copolymer material published from 2006 up until the time of writing. We neglect papers where NMR has only been used to characterize the results of chemical synthesis. Some earlier work has also been included when needed in order to put recent work into perspective. It should be remarked that investigations of self-associating block copolymers often use a multi-experimental approach, where several physico-chemical techniques are used. Here we concentrate on the NMR technique.

We have divided this review into four sections. In the first we describe general physical chemistry studies of block copolymers. The second section deals with "novel systems" while in the third section we treat studies pertaining to systems designed for drug delivery purposes. Finally, two papers which cannot be characterized within this general scheme will be dealt with in the last section.

2. General physical chemical studies

An often studied amphiphilic block copolymer is the triblock poly (ethylene oxide)-poly(propylene oxide)-poly(ethylene oxide) (PEO-PPO-PEO) system. Often it is referred to by the BASF trade name Pluronics. On account of the hydrophobicity of the PPO block, it self-aggregates in water (a recent review is). This block copolymer continues to attract attention also during the period reviewed here.

Liang et al. report relaxation time distributions from dynamic light scattering and proton NMR spectra of two Pluronics (P103, $EO_{17}PPO_{60}EO_{17}$, and P104, $EO_{27}PPO_{61}EO_{27}$) as cycled through the micellization by changing temperature. The two methods present conflicting views of the aggregation and disaggregation processes . While the NMR spectra display no hysteresis, the DLS results show a clear hysteresis effect. The reason for this is discussed in terms of the presence of a few highly swollen micelles in addition to unimers upon cooling. These aggregates must have a very long life-time (on the order of hours as each sample was allowed to equilibrate for 2 h before each measurement). Why the effect is not seen in NMR remains unclear. Could it be that the polydispersity of the Pluronics plays a role? We return to this question below.

In the micellization of Pluronic P105 ($EO_{37}PPO_{56}EO_{37}$) is investigated. The authors combine NMR based chemical shifts, spin-lattice relaxation rates and self-diffusion data to study the micellization

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in detail. The micellization appears to occur over a temperature range of around 15 °C. The spin-lattice rates for the EO and PPO parts differ. For the former, there is an exponential decrease with temperature and there is no influence by the aggregation process indicating that the segmental motion of the EO part is largely isotropic and thus to a large extent unaffected by the association. For the PPO forming the core this is not the case; here the slow motion on account of the micellar tumbling affects the relaxation rates. The diffusion data follows the same trend, although it must be said that the authors do not discuss whether the PFG data are non-linear when represented in Stejskal–Tanner plots. For these kinds of systems this is important information. In addition, it appears that the authors have used the solution viscosity in obtaining the hydrodynamic radii presented in Fig. 6.

Pede et al. present a study on the association of F68 ($EO_{76}PPO_{29}EO_{76}$) while two studies are concerned with the influence of KCl on the micellization of F88 ($EO_{103}PPO_{39}EO_{103}$) . It is shown that the salting-out ion Cl^- promotes the formation of micelles, as indicated e.g. by the lowering of the critical micell mperature (cmt). In the same vein Nilsson et al. present results based on NMR diffusometry for a number of both salting-in and salting-out ions. They present the following Hofmeister series: $SCN^- > I^- > Br^- \approx Cl^- > F^- \approx OH^-$ (where SCN^- has highest propensity to destabilize micelles). Often, the effects are ascribed to the influence the ions have on the water structure (terms such as "structure-breaking" or "structure-making" ions are often used), but in our opinion a better alternative is to describe the effects of salts in terms of their affinities (or lack of) to surfaces (see discussion in).

The paper by Nilsson et al. presents a very thorough NMR diffusometry study of the micellization of F127, $EO_{97}PPO_{68}EO_{97}$. In the pulsed field gradient experiments, the Stejskal–Tanner plots have considerable curvature in the temperature interval where micelles are formed. Below and above the temperature range, the plots are nearly linear. This was explained as being caused by the polydispersity of the Pluronics, such that at a given temperature, the more hydrophobic component is aggregated, while the more hydrophilic components remain as unimers in solution. In effect, this is a fractionation of the polymer on the basis of hydrophobicity. As the temperature changes, the composition of the micelle (and consequently of the unimer population) is altered. In fact, by the use of the CORE approach , Nilsson et al. could estimate the composition in terms of the ratio of the EO to PPO units of the micelles (and hence also for the unimers) as a function of temperature. The results are shown in

One important conclusion from the work of Nilsson et al. is the need to examine the curvature of the Stejskal–Tanner plots in NMR

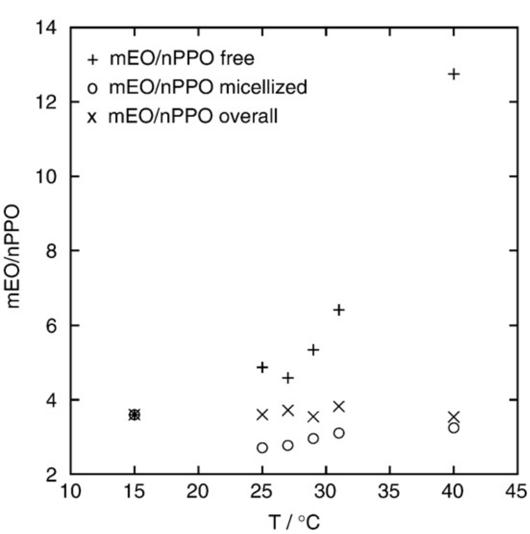


Fig. 1. Fractionation of F127 with respect to hydrophobicity during micellization. The ratio between the numbers of EO units (m) to PO units (n) versus T. The cases of free, micellized, and overall are shown.

diffusometry experiments from Pluronics and related systems. This requires access to accurate pulsed-gradient spin-echo data covering a large range of magnetic field gradient strengths.

Pluronics, being synthetic polymeric molecules, are polydisperse both in overall size and in the ratio of the block sizes, and this has to be remembered in any analysis of experimental data. In addition, there may be diblocks of high hydrophobicity present. Hvidt and co-workers have investigated the amount of impurities (in particular, they report that as much as 20% of the polymer is the PEO-PPO diblock) as well as the influence of the polydispersity on the micellization process . In the present reviewers' opinion, this may very well explain the indication in light scattering studies of (very) long-lived aggregates. To make matters worse, the polydispersity of a given Pluronics is most likely batch dependence, a fact which may explain the often cited observation that cmc and cmt values are not reproducible.

There are a number of studies on the influence of additives on Pluronics. The effect of n-alkanols on the aggregation of Pluronic P123 is described . While short chain alcohols increase both the cloud point and the cmt, longer chain alkanols induce an opposite effect. This is shown using a variety of techniques, including proton NMR where the broadening of the polymer proton signals an increase in the hydrodynamic diameter of the studied species due to aggregation. The short chain alkanols decrease the selectivity of the solvent. The long chain ones, on the other hand, co-aggregate with the polymer, thus making the polymer effectively more hydrophobic and as a consequence decreasing the cmt. The solubilization of the polar oil tributylphosphate (TBP) in micelles of a number of Pluronics has been studied . Based on a detailed analysis of the evolution of proton spectra of the amphiphilic polymers as the amount of TBP is increased, the authors conclude that the hydration state of the PPO core is altered. Little effect is seen on the EO part. As the amount of TBP is increased, there is an associated change in the solution microstructure from spherical swelled micelles to nanodroplets of pure TBP stabilized by the Pluronic molecules. Related is a study on the influence of an ionic liquid, 1-butyl-3-methyl-imidazolium bromide, on aggregation behavior of Pluronics 104 . Selective NOE data indicate that the PPO block interacts with the butyl group of the ionic liquid.

In an interesting application of several novel NMR techniques, Douglass et al. report a study on wormlike micelles , which are induced from spherical P105 micelles by the addition of 1-phenylethanol-d_5. They combine rheology and various NMR techniques including spatially resolved NMR spectroscopy and velocimetry. A variety of shear-banding and alignment behaviors are observed, along with both stable and fluctuating flows. The system shows a very rich rheological landscape; considerably more complex than what is found for the more classical wormlike micelles formed from low molecular weight surfactants. The presented techniques hold great promise for the investigation of the often complex rheology shown in associating polymer solutions; this complexity and the ability to tune the flow properties often forms the basis for their applications.

Turning to studies of block copolymer systems other than Pluronics, Bernaerts et al. investigate diblock copolymers formed by poly(tetrahydrofuran-*b*-*tert*-butyl acrylate) (PTHF-*b*-PtBA) and poly(tetrahydrofuran-*b*-1-ethoxyethyl acrylate) (PTHF-*b*-PEEA) . Hydrolysis or thermolysis of the diblock copolymers results in amphiphilic pH responsive copolymers PTHF-*b*-poly(acrylic acid) (PTHF-*b*-PAA). These form micelles at high pH values, the properties of which can be adjusted by changing pH.

A report on a novel amphiphilic thermosensitive star copolymer with a hydrophobic hyperbranched poly (3-ethyl-3-(hydroxymethyl) oxetane) (HBPO) core and many hydrophilic poly(2-(dimethylamino) ethyl methacrylate) (PDMAEMA) arms is presented . The polymer was used as the precursor for aqueous solution self-assembly. When the solvent is made selective for the hydrophilic arms, the NMR peaks from the core broadens as core-shell particles are formed. AB block copolymers of 4-vinylbenzyltrimethylphosphonium chloride (TMP) and *N*,*N*-dimethylbenzylvinylamine (DMBVA) are produced by reversible addition-fragmentation chain transfer (RAFT) radical polymerization . The pH dependent aggregation is described, again using the fact that the NMR signals from the core broaden. A related study is presented in .

Worm-like aggregates with a PAA/P4VP complex core and a PEG/PNIPAM mixed shell were prepared in ethanol by the comicellization of poly(ethylene glycol)-block-poly(acrylic acid) (PEG-*b*-PAA) and poly(*N*-isopropylacrylamide)-block-poly(4-vinylpyridine) (PNIPAM-*b*-P4VP) . This study uses broadening of NMR signals from the micellar core to indicate aggregation. A novel double-hydrophilic block copolymer poly(*N*-isopropylacrylamide)-block-poly(4-vinylpyridine) (PNIPAM-*b*-P4VP) with low polydispersity which could respond to both temperature and pH stimuli in aqueous solution was synthesized by atom transfer radical polymerization . Here also the broadening of signals from core is used to indicate aggregation. Yuan et al. describe a system of diblock copolymer micelles comprising cationic poly(2-(dimethylamino)ethyl methacrylate) (PDMA) coronas and hydrophobic poly(2-(diisopropylamino)ethyl methacrylate) (PDPA) cores are used as nanosized templates for the deposition of silica from aqueous solution at pH 7.2 and 20 °C. NMR is used to monitor the protonation state of hydrophobic moiety. Upon micellization and the formation of a hydrophobic core, deprotonation occurs.

Amino acid derived norbornene monomers, *N*,*N*′-(*exo*-bicyclo[2.2.1]hept-5-en-2,3-diyldicarbonyl)bis-L-leucine methyl ester and *N*,*N*′-(*exo*-bicyclo[2.2.1]hept-5-en-2,3-diyldicarbonyl) bis-L-leucine were polymerized by Sutthasupa et al. . Homopolymers as well as random and block copolymers were obtained in good yields. When immersed in acetone they form reversed micelles with a hydrophilic core and a hydrophobic corona, as evidenced by line broadening of the proton NMR signals from the core.

A doubly hydrophilic diblock copolymer is investigated . The polymer contains a pH sensitive poly(hexa-(ethylene glycol) methacrylate) (PHEGMA), block and an ionizable poly(2-(diethylamino)ethyl methacrylate) (PDEAEMA), block. The ionization can conveniently be followed from the proton NMR spectra. At high degree of ionization the polymer is present as unimers while at a degree of ionization of around 0.2, aggregation commences. To form proper micelles complete deprotonation is required, in order for the PDAEMA block to form a non-polar micellar core.

A mixture of two biocompatible polymers, poly(ethylene glycol)-*b*-poly(ε-caprolactone) (PEG5000-*b*-PCLx) and 1,2-distearoyl-*sn*-glycero-3-phosphoethanolamine-*N*-methoxy poly(ethylene glycol) (PEGDSPE) is described . Mixed micelles are formed with a hydrophobic core and a stabilizing corona of poly(ethylene glycol). NOESY studies indicate that the polymers together form a homogenous core. Some penetration of the PEG segments is also indicated. We end this section with a report on mixtures of poly(oxyethylene)-polydimethylsiloxane diblock polymers $Si_mC_3EO_n$, and a short chain nonionic surfactant through NMR self-diffusion data . For short silicone chains ($m=5.8$) the polymer and surfactant mix in the micelles at all mixing ratios. As the silicone chain is made longer, there is coexistence of two kinds of micelles, differing in the mixing ratio of the two components. From the data, the composition of the micelles is obtained. The two kinds of micelles differ in size; the polymer resides exclusively on the large ones while the surfactant is found in both small and large micelles. This work constitutes an illustrative example of the use of NMR diffusometry to obtain in a straightforward way non-trivial information about self-aggregated systems.

3. Novel systems

A number of new types of amphiphilic molecules with a block copolymer structure have been synthesized for potential use in drug delivery systems, in cosmetics or other applications. As noted above, one of the blocks is soluble in one type of solvent while the other block is soluble in another solvent. In this way, one can make aggregates or micelles with the soluble part of a number of polymeric molecules forming a corona and the insoluble block forming the core, by carefully choosing the solvent. One can also utilize the response of the various blocks' solubility to changes of the solvent's quality, by changing the temperature, or by e.g. introducing ions (salt) in aqueous solutions. Ionizable groups present in one block can be responsive to changes in pH, making this more or less soluble compared to the other block. Different responses for the blocks to temperature of the solvent have been mentioned previously, and also this effect may be used to control micellization of certain types of block copolymers (see above). Indeed, there seems to be many ways to manipulate block copolymers to form aggregates of various architecture for special applications — only ones fantasy limits the possibilities.

A lot of experimental techniques are used to characterize both synthesis products, physical properties like cmc and cmt, and micellar aggregates formed by the block copolymers. These techniques include static and dynamic light scattering (DLS), differential scanning calorimetry (DSC), IR, UV–vis and fluorescence spectroscopy, and NMR methods. In general, the NMR methods used are quite simple. ^{1}H and ^{13}C spectroscopy are used in a qualitative way to indicate what block of the molecules forms the corona and what block forms the core. Other, more advanced applications of NMR, like pulsed field gradient NMR to monitor the self-diffusion of the components, or spin relaxation measurements to more quantitatively determine the dynamic state of the aggregated molecules, seem to be rather scarce.

Bahadur et al. present a synthesis of a biodegradable triblock copolymer of the form poly(*p*-dioxanone-co-caprolactone)-block-poly(ethylene oxide)-block-poly(*p*-dioxanone-co-caprolactone) with molecular weights (number average, M_n) in a range around 40 000 Da with a polydispersity index in the range of 1–2, and with varying weight ratios caprolactone:*p*-dioxanone in the two end blocks. A random mix of caprolactone and *p*-dioxanone in the blocks is demonstrated using ^{1}H NMR spectroscopy. cmc values in the range of 0.8–2 mg/L in an aqueous solution buffered to a pH of 7.4 and at room temperature, were determined using fluorescence methods. Micellar aggregates with a spherical shape have been demonstrated to exist, using atomic force microscopy (AFM). A hydrodynamic diameter for the micellar aggregates in the range ca 90 to 110 nm has been determined using DLS. Interestingly, AFM shows aggregate diameters that are only 50% or less than those calculated from DLS measurements, but the present reviewers cannot see that the authors explain this discrepancy The end blocks form the hydrophobic core while the poly(ethylene oxide) block forms the hydrophilic corona at room temperature. In a qualitative way this is seen in ^{1}H spectra of micellized solutions, in that proton signals from the groups of the outer blocks get very broad and not detectable compared to the situation when the polymers are dissolved in deutero-chloroform ($CDCl_3$) where the polymers do not aggregate. This effect is demonstrated in Fig. 7 of that work.

Chang and Dai have synthesized a new class of dendrimers consisting of hydrophobic oligo(*p*-phenylene vinylene) core branches and hydrophilic oligo(ethylene oxide) chains. The motivation for the synthesis of this kind of amphipilic compounds is to form systems that respond to certain external stimuli, like changes in temperature, pH, etc in a controllable and reversible way that can be utilized in e.g. sensor applications. The hydrophobic part of these molecules in addition has some photophysical properties that can be utilized in this connection, in that temperature- and concentration dependent photophysical properties (e.g. UV–vis) have been demonstrated. The aqueous solutions show a lower critical solution temperature behaviour, due to the oligo (ethylene oxide) chains that respond to temperature in this way. cmc values have been determined at 25 °C, using surface tension measurements, to be in the range of 0.6–4.0 mg/L, increasing with the number of "arms" in the molecules from two to four. Aggregation numbers are not measured, and no estimates are given regarding the degree of cooperativity that is operating in these systems. For a more complete understanding of the thermodynamic behavior information of this kind is wanted.

[1]H NMR has been used also in this study to qualitatively assess the dynamic state of the various parts of the molecule as the micellization process occurs. This behavior is summarized in two figures in . One (Fig. 2 in) shows the difference between the proton NMR spectrum of one of the three-armed surfactant in deuterochloroform and in heavy water. While all signals are well resolved in chloroform, which is a good solvent for both kinds of blocks, the aromatic protons give rise to very broad signals in water. The aliphatic protons in the oligo(ethylene oxide) chains are still narrow. The other figure (Fig. 3 in) shows the gradual broadening of the signals originating from the aromatic protons in the three-armed molecules as the temperature is increased. This phenomenon is interpreted as a decreased local mobility of the aromatic part as the temperature is increased, due to a gradually decreased solubility of the hydrophilic groups that induce a more rigid shell as the lower critical solution temperature is approached.

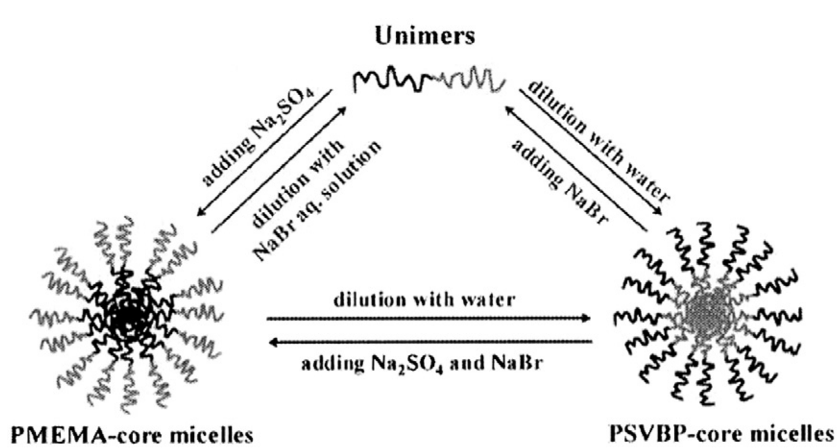
This same way to interpret broadening of proton NMR signals from groups forming a micellar core in a selective solvent (water), is used in several papers. In a work by Lee et al. copolymers consisting of a hydrophobic poly(L-lactide) (PLLA) block with grafted chondroitin sulfate (CS) are synthesized. These types of molecules may have potential for use in drug delivery applications.

Critical aggregation concentration (cac) is found to be in the range of 4 to 9 mg/L at 25 °C, depending on the composition of the molecules. The diameter of the aggregates, as determined by DLS, is in the range of 90–180 nm, depending on the degree of polymerization of L-lactide units. These findings are summarized in Table 1 of that work. The aggregates form a core of PLLA with (ionized) CS groups forming a corona. This is again manifest in proton NMR spectra of polymers dissolved in deuterated dimethyl sulfoxide (d_6-DMSO) and in heavy water, respectively. In DMSO NMR signals from protons in the PLLA part of the molecules are present in the spectra, while in heavy water, they disappear or become very broad. Again, no information on possible cooperativity of a micellization process is given – does the aggregation process involve more than single molecules that undergo some kind of helix – random coil transition?

The same comment may also be made about the work by Zhang and Guo . They have synthesized triblock amphiphilic copolymers consisting of two hydrophobic poly(sebacic anhydride) (PSA) blocks and one central hydrophilic poly(ethylene oxide) (PEO) block. By a precipitation/solvent evaporation process, the polymers are dissolved in water, and by comparing [1]H NMR signals from groups belonging to the two types of blocks when the polymers are dissolved in chloroform and in water, the NMR signals from the PSA block disappear from the spectra in the latter case — again indicating that these blocks form a core in micelles that are assumed to have some flower-type structure (Fig. 8 of their work shows a drawing of the micelles). Cmc values are not given and are probably not measured. Hydrodynamic diameters, measured using DLS, are found to depend on the size of the PSA blocks, keeping the PEO block size constant. In itself, this is not surprising, but Fig. 10 of shows a steady increase in this diameter with increasing polymer concentration. In addition, as summarized in Table 4 of , a substantial growth of aggregates seems to occur when increasing the rate of dropping a solution of the polymer in tetrahydrofuran (THF) into water during the precipitation/solvent evaporation procedure. This effect indicates that the systems are not at equilibrium. When using different solvents like acetonitrile, THF, and acetone, also different aggregate sizes formed in water are found. If the solvent is completely evaporated, and the concentration of polymer in the resulting aqueous solution is the same, one should indeed expect the resulting structures to be identical, if the systems are at thermodynamic equilibrium.

Wang et al. have synthesized a sulfobetaine diblock copolymer, of the type poly(*N*-(morpholino)ethyl methacrylate)-*b*-poly(4-(2-sulfoethyl)-1-(4-vinylbenzyl)pyridinium betaine) (PMEMA-*b*-PSVPB). Ca 30 monomer units in both blocks give a weight average molecular weight (M_w) of 54000 Da, determined by light scattering. By changing the salt composition of the aqueous solutions, it is possible to invert the microstructure of the micelles due to the fact that the two blocks show different solubility in salt free water and in aqueous solutions with certain concentrations of sodium bromide and sodium sulfate. This effect is illustrated in below which is a reproduction of part (b) of Scheme 1 of this work. Experimentally, this effect is demonstrated in [1]H NMR spectra of solutions containing the polymers as unimers and at conditions where the two types of micelles are formed (in). Cooperativity of the micellization process is in this work well documented, and summarized in Table 1 in . In addition, the kinetics of the micellization processes are studied, utilizing a stopped-flow technique. A characteristic relaxation time for the formation of PMEMA core micelles in aqueous solutions of sodium sulfate and sodium bromide, is e.g. found to be in the range 0.3–0.6 s. Other micellar processes are found also to be in the second-regime in these systems.

Ogura et al. have synthesized a thermoresponsive diblock copolymer consisting of a hydrophobic block of poly(ethyl glycidylether) and a hydrophilic block of poly(ethylene oxide), in short PEGE-*b*-PEO. Molecular weights span the range ca 3000 to 10000 Da, with varying length of the hydrophobic block, keeping the PEO block



. Schematic illustration of the micellization behavior of the PMEMA-*b*-PSVBP diblock copolymer in aqueous solution at different conditions. Reprinted with permission from Copyright 2007 American Chemical Society.

constant at 93 EO units. ^{1}H NMR spectra of a 1 wt.% aqueous polymer solution as a function of temperature show that the PEGE block forms a core and the PEO block forms a hydrated corona in micelles that are formed at temperatures above of 15 °C. No indication is given regarding a lower concentration for possible micelle formation. In this study, however, pulsed field gradient NMR (PGSE-NMR) is used to determine hydrodynamic radii of the aggregates based on the self-diffusion coefficients of the polymer molecules. Above the cmt of ca 15 °C the micelles have a hydrodynamic radius in the range of 10–15 nm. Above 40 °C a further growth of the hydrodynamic radius is demonstrated using both PGSE-NMR and DLS measurements. This phenomenon is ascribed to a further aggregation of the core-shell micelles, and in our opinion probably reflects a phase separation that eventually takes place upon dehydration of the PEO chain with increasing temperature. In a phase diagram (their Fig. 2) an opaque sol is indicated at temperatures above ca 43 °C.

Echo attenuation plots for a 1 wt.% aqueous solution as a function of temperature for one of the polymers are shown in Fig. 5 of . For systems where one self-diffusion coefficient describes the self-diffusion process plots like these give straight lines, due to a single exponential decay. However, in several of the echo attenuation plots in Fig. 5, there is a deviation from this single exponential decay in the temperature interval where micelles are formed. This behavior is interpreted as a double exponential decay, where the fast process is ascribed to unimer self-diffusion, and the slower process to the micelle self-diffusion. For this to be the case, the exchange between the monomer and micellar states of the polymer molecules must be of the order of seconds — i.e. comparable to or larger than the observation time in the PGSE experiments. It should be recalled (see above) that the non-exponential appearance of such plots is indeed quite usual in systems of this kind. The attenuation is in general not bi-exponential, but appears to be more complex, and may be ascribed to the polydispersity of the polymer molecules (see also discussion in . However, it should be recalled that the polydispersity of the molecules of the present study is small, with M_w/M_n of ca 1.2 (see Table 1 of).

4. Systems designed for drug delivery

Only three published works have been found in the period of the last two years where NMR has been used in the characterization of the physical chemistry of novel polymeric molecules with a block structure and with potential use as drug carriers. Of course, polymeric molecules that are designed for this kind of use, must be biocompatible and non-toxic. They must be able to carry a drug through rather harsh conditions like low pH, while at the same time being able to deliver the drug at the right position as a result of degradation. Also, the degradation products have to be non-toxic. The class of substances that can be used for polymerization in this context is therefore limited.

In the three works cited below, the use of NMR is mainly the standard procedure described above, namely the comparison of ^{1}H spectra of the polymers when dissolved in a good solvent for all blocks, and when dissolved in water where micellization takes place due to bad solubility of the core forming block. In qualitative terms, the formation of a core of one of the blocks is seen as a reduction/broadening of the proton signals from groups belonging to this block, compared to signals from the corona forming block or blocks.

In a work of Chen et al. a triblock copolymer with the structure poly(3-hydroxybutyrate)-poly(ethylene oxide)-poly(3-hydroxybutyrate) (PHB-PEO-PHB), has been synthesized. A range of molecular weights for the polymers has been obtained, from of 5000 to 68 000 Da, by varying the end block size from 7 to 368 3-hydroxybutyrate units, keeping the middle block constant in size with 91 EO units. Cmc values have been determined at room temperature using fluorescence methods to be in the range from 3 to 14 mg/L, decreasing with increasing length of the end blocks. Comparing proton NMR spectra of the polymers when dissolved in deuterochloroform and in heavy water, it is seen that the signals from the outer blocks get broad and weak in water compared to the signals from the groups of these blocks when the polymer is dissolved in chloroform. These blocks therefore seem to form the core of the micelles. Pyrene was used as a hydrophobic imitative drug, and using an enzyme, (extracellular) PHB polymerase, the *in vitro* biodegradation of the polymers was examined also using proton NMR spectroscopy. It was concluded that the main biodegradation products of the hydrophobic blocks were (biocompatible) 3HB monomers, dimers and trimers, with monomers as the dominating species.

Wei and coworkers have published two papers on star block copolymers that may be used as drug carriers. In the first work a four-arm star block copolymer consisting of a hydrophobic poly (methyl acrylate) (PMMA) arm and three hydrophilic poly(*N*-isopropylacrylamide) (PNIPAAm) arms was synthesized, yielding a molecular weight (M_w) of 81 000 Da, with a rather small polydispersity index, M_w/M_n, of 1.07. Comparing the proton NMR spectra obtained when the polymers are dissolved in deuterochloroform and in heavy water, the core-shell structure of micelles in water is demonstrated, with the PMMA block forming a core. A lower critical solution temperature (LCST) of 34 °C exists for this aqueous system, and a cmc value of 10 mg/L is quoted. The micelles are claimed to be of spherical shape, with a "size" of ca 50 nm, based on TEM pictures. At the LCST the PNIPAAm arms get hydrophobic, and the solutions become turbid. Also, looking at the drug release profile using the hydrophobic drug prednisone acetate, below and above the LCST, there is a difference attributed to the ability of the PNIPAAm arms to function as a hydrophobic site at temperatures above the LCST. To us, it seems somewhat speculative to relate this drug release behavior to possible changes in the micellar microstructure. The turbidity seen at temperatures above the LCST most probably indicates an aggregate growth, related to a phase separation phenomenon.

The other paper by Wei et al. deals with the synthesis of a seven-arm star block copolymer of the type poly(L-lactide-*star block*-*N*-isopropylacrylamide) (PLLA-*sb*-PNIPAAm), consisting of a hydrophobic PLLA arm and six hydrophilic PNIPAAm arms. The molecular weight (M_n) is ca 247 000 Da, with M_w/M_n of 1.6. The polymers form micelles in aqueous solution, and a cmc is found to be ca 56 mg/L, as determined using fluorescence. The hydrophobic nature of the PLLA block (arm) and the hydrophilic nature of the other block (arms) are demonstrated by comparing the proton NMR spectra of the polymers dissolved in deuterochloroform and in heavy water. The micelles have an average diameter around 100 nm, determined from TEM images. There is a size polydispersity for the micelles, according to Fig. 4b of this work. At the same time the authors claim that the micelles are of spherical shape, as evidenced in TEM observation. The size polydispersity, however, may signal the presence of micelles of a somewhat more anisometric nature. Also for this polymer, a LCST exists, and is found to be 31 °C. There is again a difference in the drug release profiles below and above the LCST observed using a hydrophobic anticancer drug (MTX). The same type of speculation regarding the micelle microstructure as in the former paper is used to rationalize the observations.

5. Other works

We have examined two works that do not belong to the categories used above. One by Joo et al. deals with the synthesis of a multiblock copolymer consisting of poly (L-lactide) and poly (ethylene oxide), (PEO-PLLA), and a corresponding copolymer of stereoisomeric poly(DL-lactide), (PEO-PDLLA). The block sizes correspond to a molecular weight (M_n) of ca 1000 Da for the PEO blocks, and ca 1500 Da for the PLLA and PDLLA blocks. M_n for the block copolymers are ca 14 000 Da, with M_w/M_n of 1.3, as determined using gel permeation chromatography (GPC). In aqueous solution at 20 °C, cmc values of $4.0 \cdot 10^{-3}$ and $7.0 \cdot 10^{-3}$ wt.%, were quoted, for the PEO-PLLA, and the PEO-PDLLA copolymer, respectively. Parallel to this difference in ability to self-assembly, it was also found differences in ability to form a gel state for aqueous solutions of the two copolymers. A gel state is formed for the PEO-PLLA system at concentrations above 8 wt.%, and for the PEO-PDLLA system, at concentration above ca 9 wt.%, in a temperature range between ca 10 and 40 °C. This behavior is depicted in Fig. 1a of . It is found that the latter polymer with an atactic arrangement of the methyl groups has a lower tendency to form a gel state than the former one with an isotactic arrangement of these groups. Using ^{13}C NMR spectroscopy a higher dynamic state, in the gel, for these methyl groups is revealed for the PDLLA based copolymer in comparison to the other copolymer.

In a contribution by Chen et al. copolymers of the type poly (ethylene oxide)-poly(propylene oxide)-poly(ethylene oxide), were used to study the effect of hydrophobicity of the polymers on the formation of gold nanoparticles. For these polymers, the PPO block forms a hydrophobic core and the PEO blocks form a hydrophilic corona in micelles at temperatures above a cmt, that depends on the molecular weight of the polymers and the relative sizes of the blocks. The ability of micelle formation is known to depend also on presence of salt, like NaF. See also Section 1 above. In this study, 5 different Pluronics with molecular weights (M_w) in the range of 5000 to 14600 Da, and with a weight % of PEO of 50 or 80, were used to examine the effect these micelle forming agents have on the formation of colloidal gold particles during the preparation of these particles, starting with $AuCl_4^-$ (aq). It is found that even at low concentration of polymer, and especially in presence of NaF, the particles attain a spherical form, as inferred from TEM images. In the absence of such polymers, the particles formed are generally non-spherical. ^{1}H NMR spectroscopy shows that the chemical shift of the methyl group of the PPO block correlates with the micelles' ability to function as a reaction medium and the polymers' ability to eventually stabilize the gold particles formed. The lowest chemical shifts are found for the P105 polymer at a salt concentration of 0.7 M. A 5 wt.% P105 solution is sufficient for the formation of spherical gold particles and the stabilization of these, at a temperature of 40 °C, as depicted in Fig. 10 in that work.

We end this review by making a general remark. In the study of core-shell particles one often makes use of the fact that the NMR signal from the core-component broadens. This is taken to indicate that micelles are formed and often discussed in terms of a dense core being formed. While the broadening is certainly an indication of aggregation, we feel that it is more accurate to explain the broadening in terms of contribution of the slow motions of the micelles. The situation is analogous to that found in the case of surfactant micelles (see e.g.); the motion in the core is anisotropic and hence a part of the relevant interaction is averaged by the slower over-all tumbling. For the shell component, the motion is closer to being isotropic and hence this part is less affected by slower motions.

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